

YUSRA AZEEM

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Summary

Software Engineer (MCA) with hands-on experience owning features end to end, from design and development to testing and deployment, across backend and cloud-hosted applications using **Java, Python, Node.js, and PostgreSQL**. Experienced in building secure **REST APIs**, automating background workflows, and optimizing performance, with strong DSA fundamentals (300+ problems) and a habit of careful code review and attention to detail.

Education

Indira Gandhi Delhi Technical University for Women

Master of Computer Applications – CGPA: **8.80**

2025 – 2027

Delhi, India

University of Allahabad

Bachelor of Science (Computer Science, Mathematics and Physics) – CGPA: **7.00**

2022 – 2025

Allahabad, India

Experience

Ransh Innovations Pvt. Ltd. | Live Demo

January 2026 – July 2026

Web Development Intern

Remote

- Migrated a production EdTech platform from Create React App to **Next.js and TypeScript**, restructuring 10+ UI elements into modular, reusable components with clear separation of concerns to simplify debugging and improve maintainability.
- Engineered and maintained **4+ production modules** (Notes, Flashcards, About, Vocabulary), integrating **REST APIs** with **Redux** and modernizing legacy component patterns for a more scalable frontend.
- Refactored component styling using **Tailwind CSS**, establishing reusable UI patterns and a consistent design system across the platform.
- Optimized performance across **5+ components** by reducing unnecessary React re-renders using **useMemo, useCallback, and React.memo** on data-heavy screens.

Projects

Algonest – AI-Powered EdTech Platform | Live Demo | GitHub

Jan 2026 – July 2026

Node.js, Express.js, React, PostgreSQL (Supabase), pgvector, Claude API (Haiku/Sonnet), Zoom API

- Owned the design, development, testing, and deployment** (hosted on **Render**) of backend services for an AI-powered EdTech platform supporting mentor-guided sessions, booking workflows, administrative approvals, and personalized learning roadmaps.
- Designed secure RESTful APIs** following **MVC architecture** with **JWT authentication** and **role-based access control (RBAC)** to protect sensitive user data, covering mentor scheduling, booking workflows, and progress tracking; validated endpoints using **Postman**.
- Implemented a persistent **RAG memory system** using Claude, PostgreSQL, and **pgvector**, summarizing student conversations into per-student memories and retrieving relevant context via vector similarity search to personalize future AI interactions.
- Built scheduled background jobs with **pg_cron** to automate mentor availability rollovers, session reminders, and cleanup of expired records, keeping data consistent without manual intervention.

Exoplanet Detection & Habitability Prediction | Live Demo | GitHub

April 2026 – June 2026

FastAPI, React, Scikit-learn, XGBoost, SHAP, Pandas

- Engineered and deployed (**Vercel**) a two-stage **FastAPI and React** application following a client-server architecture for real-time exoplanet detection and habitability prediction using XGBoost models trained on 4,000+ records.
- Developed **FastAPI REST inference APIs** serving exoplanet detection and conditional habitability prediction.
- Optimized the data preprocessing and evaluation pipeline using **SMOTETomek** to handle class imbalance, achieving **100% recall** and an **F1-score of 0.846**.
- Integrated SHAP-based model explainability to highlight key features influencing habitability predictions, improving interpretability of model outputs.

Technical Skills

- Languages:** Java, Python, JavaScript (ES6+), TypeScript, SQL
- Backend & APIs:** REST APIs, MVC Architecture, Node.js, Express.js, FastAPI, JWT Authentication, RBAC, Postman, pg_cron
- Cloud & Deployment:** AWS EC2, Render, Vercel, Git, GitHub
- Databases:** PostgreSQL (Supabase), MySQL, pgvector
- Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS
- Frontend:** React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS
- AI/ML:** Scikit-learn, XGBoost, SHAP, Pandas, RAG, Vector Embeddings

Achievements

- Solved **300+** Data Structures & Algorithms problems across **LeetCode** and other platforms, covering Graphs, Trees, Dynamic Programming, and Binary Search.
- Open Source Contributor (Snowscript):** Contributed **7 pull requests** across multiple repositories with **6 merged** after code review and feedback.
- AIWOS Start-a-thon:** Ranked **Top 250 out of 10,000+** participants nationwide.
- Secured **Rank 7** in the IGDTUW CET Entrance Test.
- Academic:** Ranked in the **top 5%** of the BSc batch at **University of Allahabad** (2025).